ESBM 3700
ENTREPRENEURIAL ENVIRONMENTS

LECTURE 6
LEAN STARTUP

University of Colorado Boulder

Our path

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Ideas to Opportunities	Choosing Your Technology	Intellectual Property Strategy
Business Models	Choosing Your Identity	Value Chain Strategy
Sources of Funding	Choosing Your Competition	Architectural Strategy
Lean Startup	Quiz	Bonus: Startup Failure
Lean Startup Exercise		Bonus: Shark Tank
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The Lean Startup

Eric Ries

Three (dangerous) approaches to launch a startup

- **Build-It-And-They-Will-Come** bypasses customer feedback and demand validation, relying solely on a founder's vision for initial guidance, and then focusing an engineering-dominated team's energy on turning the founder's vision into reality. **Wrong**

- **Waterfall Planning** divides product development work into phases (e.g., design, coding, testing) that are completed in sequence by different organizational units, with each new phase commencing only when the prior phase's work passes a formal review. **Wrong**

- **Just Do It!** eschews a strong product vision or detailed plan, relying instead upon an improvisational approach that adapts a startup's product and business model based on feedback from resource providers and customers. **Wrong**

An alternative: The lean startup approach

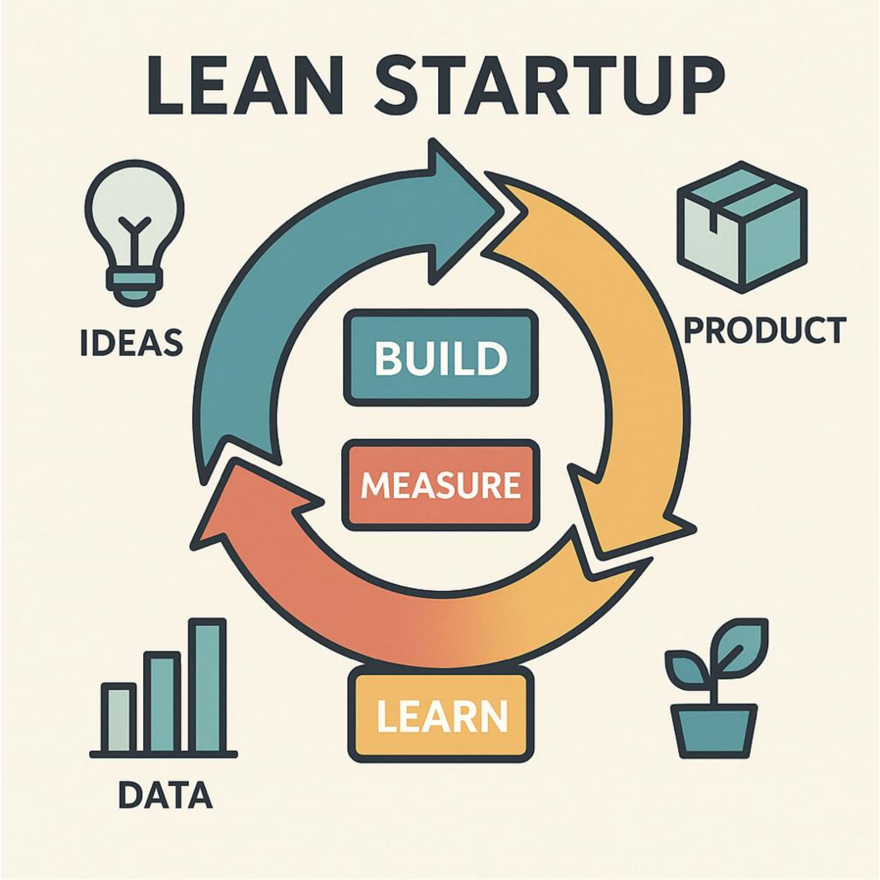
- Lean startup approach helps the entrepreneur translate her vision into *falsifiable* business model hypotheses, and then tests those hypotheses using a series of ***minimum viable products*** (MVPs).

- Based on test feedback, an entrepreneur must decide whether to ***persevere*** with her proposed business model; ***pivot*** to a revised model that changes some elements while retaining some others; or simply ***perish***, abandoning the new venture.

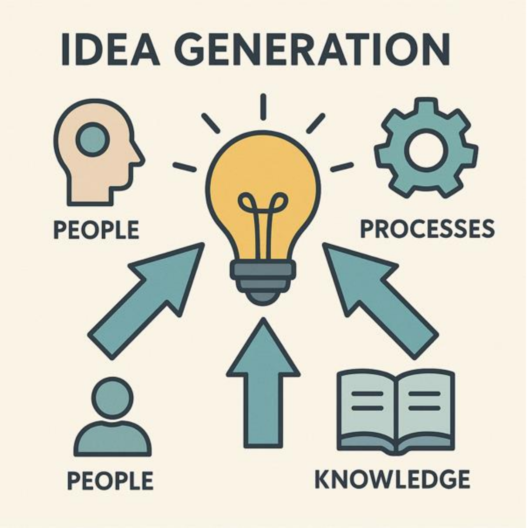
- A hypothesis-driven approach helps reduce the biggest risk facing entrepreneurs: offering a product that no one wants.

Lean startup steps

- *Step 1: Develop a vision*
- *Step 2: Translate the vision into hypotheses*
- *Step 3: Specify MVP Tests*
- *Step 4: Prioritize tests*
- *Step 5: Learn from MVP tests*
- *Step 6: Persevere, Pivot, or Perish*
- *Step 7: Scaling and ongoing optimization*



Step 1: Develop a vision



Before an entrepreneur can generate business model hypotheses, she must have a vision that her startup will address and a potential solution for that problem.

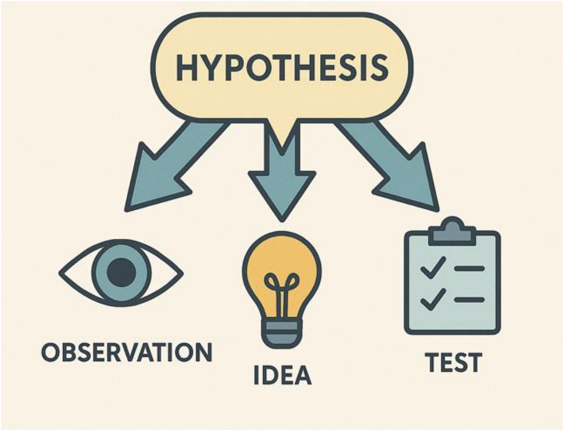
- Some ideas on ideation:
  - **Immersion:** Creativity usually follows from deep immersion in a problem.
  - **Obsession:** Creative individuals become obsessed with the problems on which they are working.
  - **Incubation:** Inventors often spend years on a problem before they get an epiphany about a solution; the subconscious remains engaged even when inventors set their work aside for long periods.
  - **Recombination:** New ideas often result from connecting seemingly unrelated concepts.
  - **Clarification:** Many inventors employ processes to keep track of and refine their ideas.
  - **Collaboration:** Most great work is done in small teams. One collaborator will say something that triggers another's ideas, and cofounders will support each other emotionally when the creative process stalls.

Step 2: Translate the vision into hypotheses

- For each business model element, an entrepreneur should formulate a set of falsifiable hypotheses.

- A hypothesis is falsifiable when it can be rejected through a decisive experiment.

- Entrepreneur should generate hypotheses that require quantitative metrics for validation.



- An example:
  - H1: Customers are willing to pay 20 cents for calls that wakes them up at a specific time.
  - H2: Customers are willing to hear advertisement when they wake up.
  - H3: Businesses are willing to pay $1 per second of advertisement over wake-up calls.
  - H4: A large share of revenue is generated between 6am-8am.

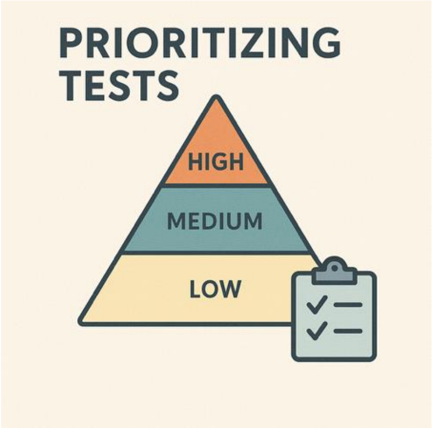
Step 3: Specify MVP Tests

- In the words of Paul Graham, Y combinator founder, entrepreneurs should "fail early and often!"
- Specify a minimum viable product (MVP), i.e., the smallest set of features and/or activities needed to test a business model hypothesis.
- Short product development batch sizes and cycle times yield two benefits:
  - **Accelerating feedback:** entrepreneurs learn about customer requirements before investing too much time in building features no one will use.
  - **Facilitating bug discovery:** releasing feature revisions in small batches makes it easier to interpret test results and to diagnose problems.
- The simplest MVPs take the form of *smoke tests* that radically constrain both functionality and operations, testing demand for a product that yet does not exist. Take Dropbox as an example!

Step 4: Prioritize tests

- As a general principle, an entrepreneur should give priority test that can eliminate considerable risk at a low cost.

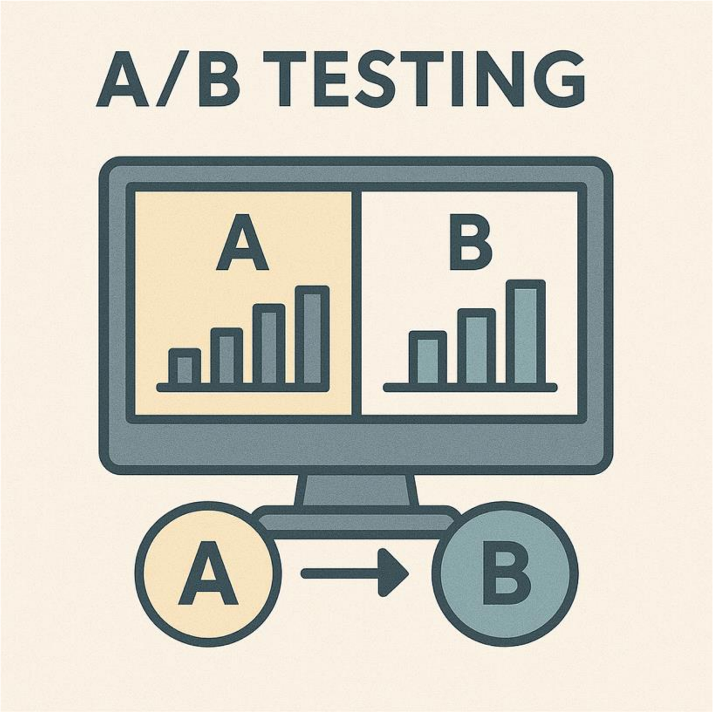
- Sometimes entrepreneurs have the option to pursue tests in parallel, because the relevant hypotheses are not serially dependent.



Step 5: Learn from MVP tests

- Gaining insights from consumer feedback might be erroneous at least for two reasons:
  - Customers' stated preferences do not always correspond to their true preferences.
  - Entrepreneur's cognitive biases can affect her evaluation of the feedback.

- The gold standard for hypothesis testing (A/B testing) is to randomly assign the objects in a sample to two groups — treatment and control. Give the MVP to the treatment group and a placebo to the control group. Observe and compare the *revealed preferences* of the two (rather than their *stated preference*).



Step 6: Persevere, Pivot, or Perish

After evaluating MVP test results and other market feedback, an entrepreneur must decide whether to persevere, pivot, or perish.

- **Persevere:** If the MVP validates the business model hypothesis and other feedback does not prompt a shift in direction, then the entrepreneur perseveres on her current path, either testing remaining hypotheses or —if all hypotheses have been validated— preparing to scale.
- **Pivot:** If the MVP test rejects the business model hypothesis, the entrepreneur may elect to pivot. A pivot changes some business model elements while retaining others, particularly core aspects of the startup’s original version.
- **Perish:** if an MVP test decisively rejects a crucial business model hypothesis, and the entrepreneur cannot identify a plausible pivot, then she shut down her business.

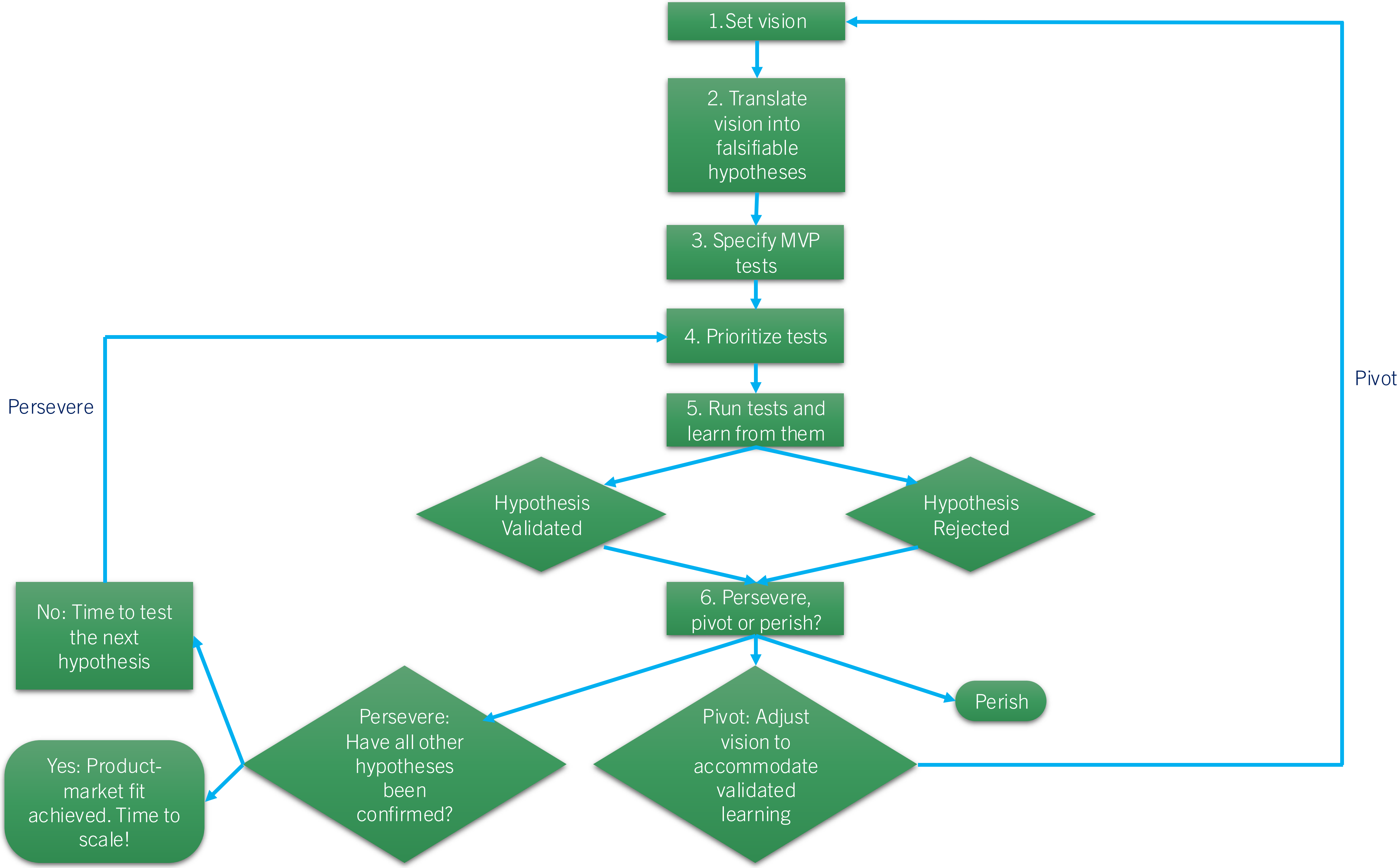
Step 7: Scaling and ongoing optimization

- If an entrepreneur has validated all key business model hypotheses, then she has achieved ***product-market fit*.**

- Product-market fit means that the venture has the right product for the market: one with demonstrated demand from early adopters and with solid profit potential.

- It is time to scale!

- Hypothesis-testing methods continue, but their focus shifts from business model *validation* to business model *optimization*.


1.Set vision
2. Translate vision into falsifiable hypotheses
3. Specify MVP tests
4. Prioritize tests
5. Run tests and learn from them
Hypothesis Validated
Hypothesis Rejected
6. Persevere, pivot or perish?
Persevere: Have all other hypotheses been confirmed?
Pivot: Adjust vision to accommodate validated learning
Perish
No: Time to test the next hypothesis
Yes: Product-market fit achieved. Time to scale!
Persevere
Pivot

THANK YOU!